

Introduction to Transfer Learning

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Video: Introduction to Transfer Learning
3 min
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Practice Quiz: Concept Check: Introduction to Transfer Learning
5 min
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Reading: Using and Comparing Pre-Trained Models
20 min
- ✓

Video: Performing Transfer Learning for Classification
6 min
- ✓

Reading: Performing Transfer Learning with Code
20 min
- ✓

Reading: Viewing Activations with the CNN Activations Explorer
10 min

Choosing Training Options

Week 2 Project: Performing Transfer Learning to Classify Traffic Signs

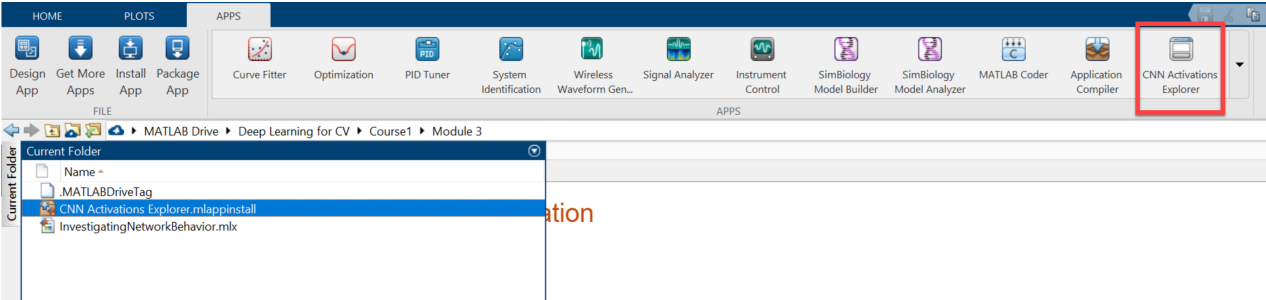
Viewing Activations with the CNN Activations Explorer

You've seen how visualizing the activations of individual layers provides insight into how CNNs perform classification.

To make it easier to explore different layers of a network and their activations for a given image, we have created a CNN Activations Explorer app. This app allows you to upload an image and choose a trained network to observe the activations for each layer.

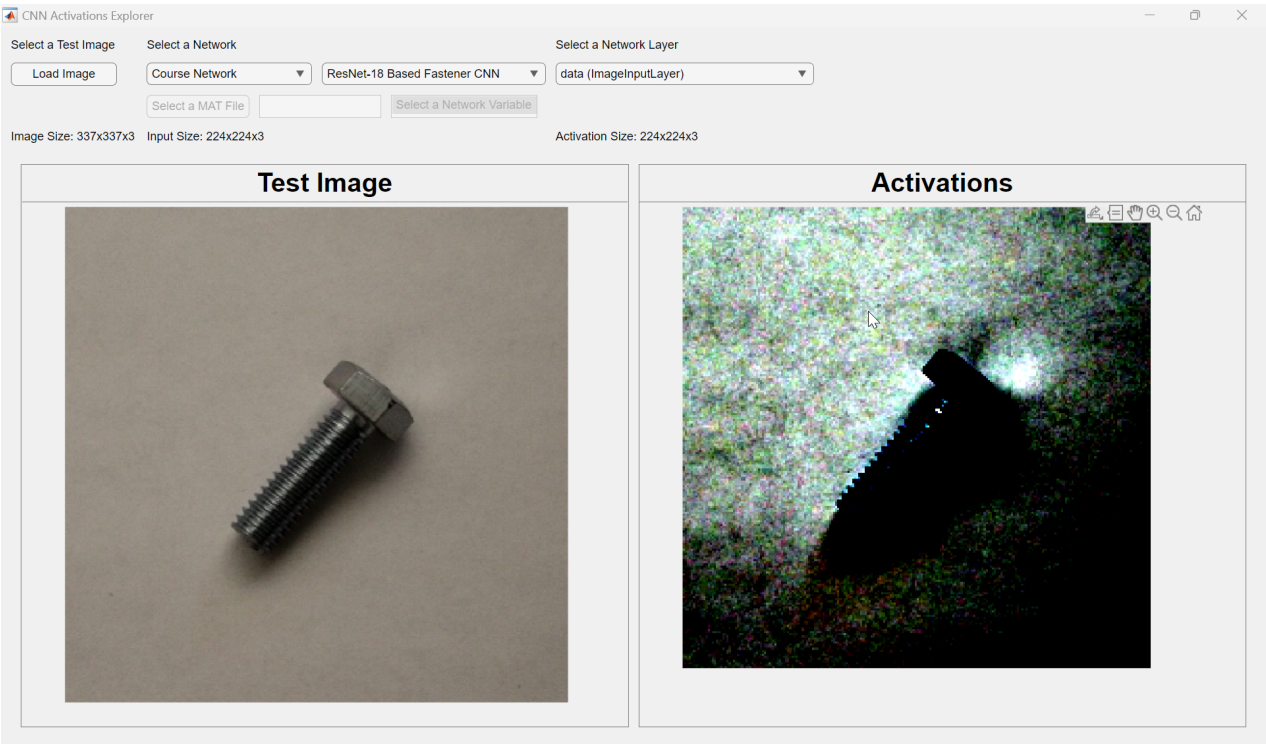
Install the app by navigating to "Course 1 > Module 2" in the Current Folder browser in MATLAB. Then double-click "CNN Activations Explorer.mlappinstall" and click "Install."

Once the app is installed, you can open it using the "Apps" tab:



Take a few minutes to experiment with the app. Load in an image and a trained network, either from the course files or a network that you have trained yourself. What do you notice? How do layers early in the network differ from downstream layers?

Notice how the image input layer activations differ from the original input:



Most image input layers perform normalization on the inputs. This wouldn't have been obvious if we hadn't thought to check the activations!

Refer to this application throughout this course to better understand how each network performs classification. Share any questions or interesting activations in the discussion forums!

Go to next item ✓ Completed

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